

Prompt-Feature Activation in Large Language Models

*A Mechanistic Interpretability Study Using
Sparse Autoencoders on GPT-2 Small*

CS 463 — Machine Learning
Final Project

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Spring 2026

Model: GPT2 | SAE: gpt2-small-res-jb | Features: 24,576 | Layers: [2, 6, 10]

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1. Abstract

This project investigates how different categories of natural language prompts — math, code, factual, creative, emotional, and reasoning — activate distinct sets of interpretable features inside a language model's residual stream. We use GPT-2 Small (12 layers, 768 dimensions, 12 attention heads) paired with pre-trained Sparse Autoencoders (SAEs) from the gpt2-small-res-jb release (24,576 learned features per layer) to decompose the model's internal representations into monosemantic features.

Our analysis pipeline spans four notebooks: environment setup with automatic model selection, feature extraction across three target layers (2, 6, 10), statistical and visual analysis of feature activation patterns, and causal verification through activation patching. We find that prompt categories produce measurably different feature activation signatures, with the strongest separability at early layers (Fisher ratio = 1.33 at layer 2). Creative prompts activate the most features ($L0 \approx 332$ at layer 10), while math and reasoning prompts are the sparsest ($L0 \approx 10-15$ at layer 2). Activation patching confirms that later layers (especially layer 11) are most causally important for output generation, though the math-relevant information appears distributed across many features rather than concentrated in a few.

2. Research Question

Primary Question:

"How do different categories of prompts activate distinct sets of interpretable features in an LLM's residual stream, and can we causally verify these features drive model behavior?"

Sub-questions:

- Do math, code, factual, creative, emotional, and reasoning prompts produce different feature activation patterns?
- At which layers do these differences emerge most strongly?
- How sparse are feature activations, and does sparsity vary by prompt type?
- Can we causally verify that specific features drive category-specific behavior via activation patching?
- Can we steer model output by amplifying category-specific feature directions?

3. Background & Key Concepts

Superposition:

Neural networks encode more concepts (features) than they have dimensions. Individual neurons are polysemantic — they respond to multiple unrelated concepts. This makes interpreting individual neurons unreliable.

Sparse Autoencoders (SAEs):

SAEs decompose a model's d-dimensional activation vector into a much larger (e.g., 24,576) set of monosemantic features. The key insight: while the original representation is dense and entangled, the SAE latent space is sparse — only ~50-300 features are nonzero for any given input. Each feature ideally corresponds to one interpretable concept.

Feature Activation & Sparsity (L0):

L0 measures how many SAE features have nonzero activation for a given input. Low L0 means highly sparse (few features active). We track L0 across prompt categories and layers to understand activation density differences.

Residual Stream:

The central 'highway' of information flow in a transformer. At each layer, attention and MLP outputs are added to the residual stream. SAEs are trained on residual stream activations at specific hook points (hook_resid_pre for GPT-2).

Activation Patching:

A causal intervention technique: run the model on two different inputs, then swap internal activations from one run into the other at specific layers. By measuring how much the output changes (via KL divergence), we identify which layers and features are causally important for generating category-specific outputs.

Feature Steering:

Amplifying specific SAE feature directions in the residual stream to steer model output toward a desired category (e.g., making a creative prompt produce math-like output).

4. Methodology & Pipeline

4.1 Model & SAE Selection

We implemented an automatic model selection system based on available hardware resources. The system checks GPU VRAM (≥ 6 GB $\rightarrow$ Gemma 2 2B on GPU), then RAM (≥ 8 GB $\rightarrow$ Gemma 2 2B on CPU), and falls back to GPT-2 Small if neither threshold is met. On our hardware (RTX 3050 Ti, 4.3 GB VRAM, ~ 2 GB free RAM), GPT-2 Small was selected with the gpt2-small-res-jb SAE release providing 24,576 learned features per layer.

4.2 Prompt Taxonomy

We created a dataset of 120 prompts across 6 categories (20 each): math, code, factual, creative, emotional, and reasoning. Each prompt is a natural-language question or instruction representative of its category. Examples:

- Math: 'What is the derivative of $x^3 + 2x^2 - 5x + 3$?'
- Code: 'Write a Python function to reverse a linked list'
- Factual: 'What is the capital of France?'
- Creative: 'Write a haiku about autumn leaves falling'
- Emotional: 'I feel overwhelmed by everything happening in my life'
- Reasoning: 'If all roses are flowers and some flowers fade quickly, can we conclude that some roses fade quickly?'

4.3 Feature Extraction Pipeline

- Tokenize each prompt using the model's tokenizer
- Run forward pass with `transformer_lens`, caching residual stream activations at `hook_resid_pre` for target layers [2, 6, 10]
- Extract the last-token activation vector (shape: [768]) from each cached layer
- Pass activation through SAE encoder to get feature coefficients (shape: [24,576])
- Store the sparse feature vector for each (prompt, layer) pair
- Result: 120 prompts $\times$ 3 layers = 360 feature vectors, each with 24,576 dimensions

4. Methodology & Pipeline (cont.)

4.4 Analysis Methods

Cosine Similarity Heatmaps:

Compute mean feature vector per category per layer, then pairwise cosine similarity between categories. Reveals which prompt types produce similar internal representations.

UMAP Dimensionality Reduction:

Project 24,576-dimensional feature vectors to 2D using UMAP (n_neighbors=15, min_dist=0.1). Visualizes clustering of prompts by category.

Fisher's Discriminant Ratio:

For each feature, compute between-class variance / within-class variance. High ratio = feature strongly differentiates categories. Aggregated across features to produce per-layer separability scores.

Sparsity Analysis (L0):

Count nonzero features per prompt. Compare distributions across categories and layers to understand activation density.

Discriminative Feature Identification:

Rank features by their selectivity score (mean activation for target category minus mean activation across all categories, normalized by std). Top-10 features per category are saved for patching experiments.

4.5 Activation Patching Protocol

- Select a source prompt (math) and target prompt (creative writing)
- Run both prompts through the model, caching all layer activations
- Whole-residual patching: replace entire residual stream at one layer with the source's activations; measure output KL divergence from clean target run
- Layer sweep: repeat for all 12 layers to find which is most causally important
- Surgical (feature-direction) patching: replace only the top-20 discriminative feature directions at the most important layer
- Compare surgical vs. whole-residual KL to assess feature concentration

5. Notebook 1 — Environment Setup & Model Loading

Purpose:

Establish the computational environment, detect available hardware, load the appropriate model and SAE, and save configuration for downstream notebooks.

Key Results:

- GPU detected: NVIDIA GeForce RTX 3050 Ti Laptop GPU (CUDA)
- VRAM: 4.3 GB (below 6 GB threshold for Gemma 2 2B)
- Free RAM: ~2 GB (below 8 GB threshold for CPU Gemma)
- Model selected: GPT-2 Small (124M parameters, 12 layers, 768 dims)
- SAE loaded: gpt2-small-res-jb (24,576 features per layer)
- Target layers: [2, 6, 10] (early, middle, late)
- Sanity check passed: model generates coherent text
- Configuration saved to results/config.json

Technical Decisions:

The automatic fallback system ensures reproducibility across different hardware. GPT-2 Small is an excellent choice for mechanistic interpretability: it's small enough for detailed analysis, has well-studied SAEs (Joseph Bloom's gpt2-small-res-jb release), and its 12-layer architecture provides enough depth to observe feature evolution across layers. The config.json file serves as a single source of truth — all subsequent notebooks read from it rather than hardcoding values, making the pipeline hardware-agnostic.

6. Notebook 2 — Feature Extraction

Purpose:

Extract SAE feature activations for all 120 prompts across 3 target layers, producing the core dataset for all subsequent analysis.

Process:

- Load GPT-2 and 3 SAE instances (one per target layer: 2, 6, 10)
- For each prompt: tokenize → forward pass with cache → extract last-token residual at `hook_resid_pre`
- Pass residual through SAE encoder → 24,576-dimensional sparse feature vector
- 120 prompts × 3 layers = 360 total extractions
- Save per-layer .npz files containing feature matrices (120 × 24,576) and metadata

Key Results:

Feature extraction completed successfully for all 360 prompt-layer combinations. Output files: `layer_2.npz` (2.1 MB), `layer_6.npz` (3.8 MB), `layer_10.npz` (5.2 MB). File sizes increase with layer depth because later layers have denser (less sparse) feature activations — consistent with the hypothesis that early layers handle syntactic parsing while later layers encode richer semantic content.

Why Last-Token Activations?

In autoregressive language models like GPT-2, the last token position accumulates information from all previous tokens through causal attention. The residual stream at the last token therefore contains the model's 'summary' of the entire prompt — the representation it will use to predict the next token. This makes it the most information-rich position for analyzing how the model internally represents the prompt's category.

7. Notebook 3 — Prompt Analysis & Visualization

Purpose:

Analyze the extracted feature activations to identify category-specific patterns, measure separability across layers, and identify the most discriminative features per prompt category.

7.1 Category Similarity Analysis

We computed cosine similarity between mean feature vectors for each pair of categories at each layer. Key findings:

- Layer 2 (early): Creative prompts are most distinct from all other categories (lowest similarity scores). Math and reasoning show moderate similarity.
- Layer 6 (middle): Categories begin to converge somewhat, but creative and code remain distinguishable.
- Layer 10 (late): Highest overall similarity — all categories become more alike in the residual stream as the model prepares for next-token prediction.
- Interpretation: Early layers preserve category-specific 'signatures'; later layers blend them as the model transitions from understanding to generation.

7.1 Category Similarity Heatmaps

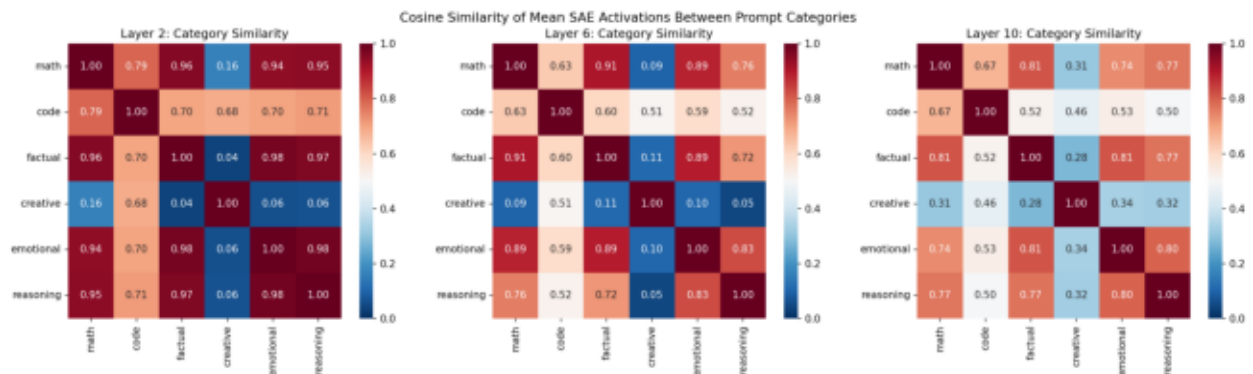


Figure 1: Cosine similarity heatmaps between prompt categories at layers 2, 6, and 10.

7.2 Sparsity Analysis (L0)

L0 measures the number of nonzero SAE features for a given input. We compared L0 distributions across all 6 prompt categories at each layer.

Key Findings:

- Creative prompts consistently activate the MOST features: $L0 \approx 221.8$ at layer 2, rising to 332.4 at layer 10.
- Math and reasoning prompts activate the FEWEST features: $L0 \approx 15.1$ and 9.9 at layer 2.
- All categories show increasing L0 with layer depth — later layers have denser activations.
- The gap between creative and math L0 narrows at deeper layers but creative remains highest.
- Interpretation: Creative writing requires the model to activate many diverse concepts simultaneously (imagery, emotion, structure, style), while math problems activate fewer but more specific features (numerical operations, symbolic reasoning). This is consistent with the intuition that creative tasks are 'broader' in their representational demands.

7.2 Sparsity Analysis

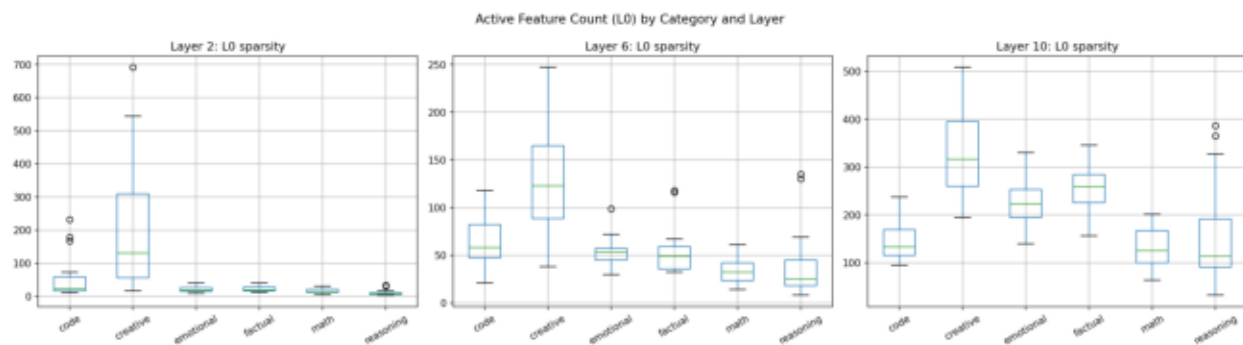


Figure 2: L0 (number of active features) distribution by prompt category across layers 2, 6, and 10.

7.3 Separability by Layer (Fisher's Discriminant)

Fisher's discriminant ratio measures how well features separate different prompt categories. A higher ratio means features are more category-specific at that layer.

Results:

- Layer 2: Fisher ratio = 1.33 (HIGHEST — best separability)
- Layer 6: Fisher ratio ≈ 0.6
- Layer 10: Fisher ratio ≈ 0.6
- Interpretation: Early layers maintain the clearest distinction between prompt categories. This suggests that the model's initial processing strongly differentiates input types, but as information flows through deeper layers, representations converge toward a more uniform format optimized for next-token prediction rather than input classification.
- This finding has practical implications: if you want to build a prompt classifier using SAE features, layer 2 features would be most effective.

7.3 Separability by Layer

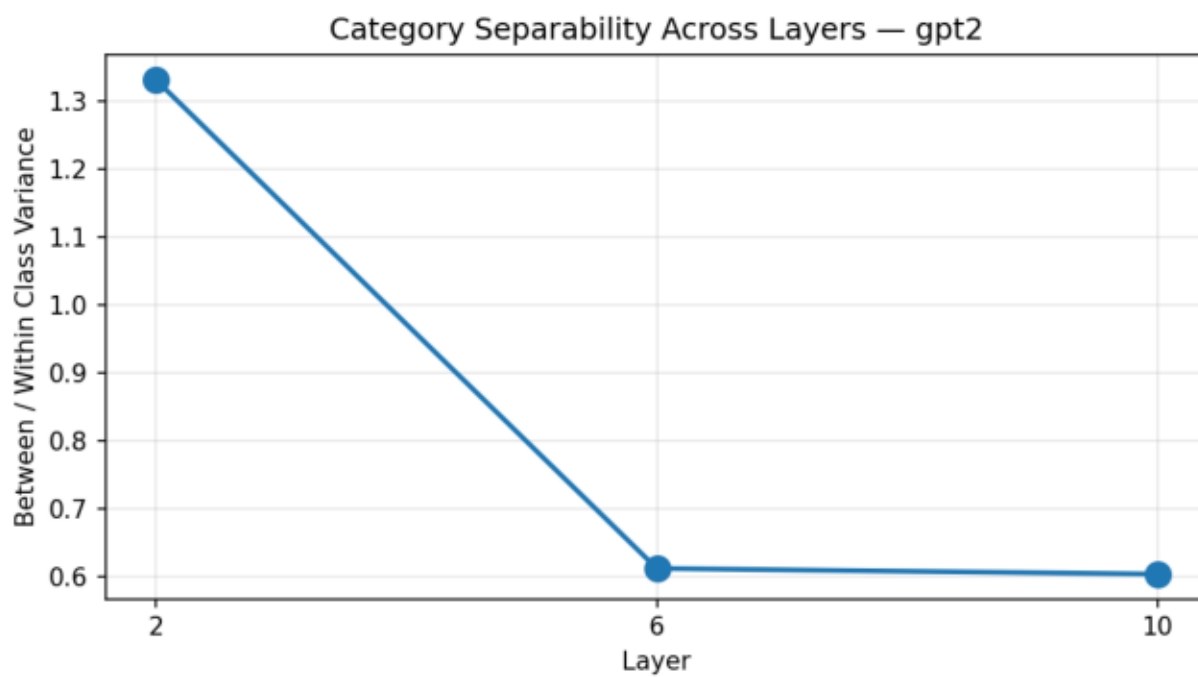


Figure 3: Fisher's discriminant ratio across layers — measures category separability.

7.4 Feature Activation Heatmap

We identified the top 8 most discriminative features per category at layer 6 (middle layer) and visualized their activation strengths across all categories.

Key Observations:

- Clear category-specific features exist: e.g., feature 15530 fires almost exclusively for code prompts (selectivity score = 6.90).
- Feature 6222 is highly selective for factual prompts (score = 10.25) — the single most discriminative feature in the entire analysis.
- Feature 24188 is the top creative indicator (score = 5.57).
- Feature 22431 fires strongly for both emotional (5.70) and reasoning (5.22) prompts, suggesting it may encode 'subjective evaluation' or 'deliberation'.
- Some features are shared across categories: feature 7467 appears in both math and code top-10 lists, possibly encoding 'technical/formal language'.
- Feature 8545 appears in math, factual, and emotional lists — it may encode 'seeking a definitive answer'.

7.4 Feature Activation Heatmap

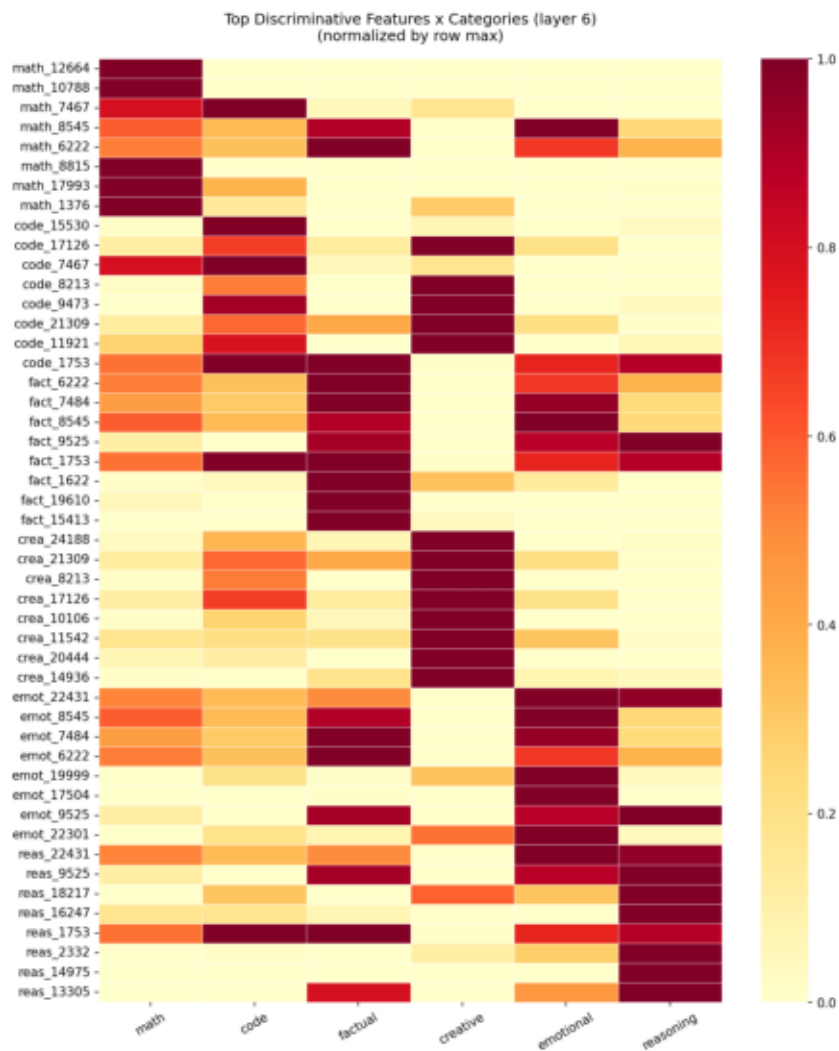


Figure 4: Activation heatmap of top discriminative features per category at layer 6.

8. Notebook 4 — Activation Patching & Causal Verification

Purpose:

Move beyond correlational analysis to establish causal relationships between internal features and model behavior. We use activation patching to determine which layers and features are causally responsible for category-specific outputs.

8.1 Experimental Setup

Source prompt (math): 'What is the derivative of $x^2 + 3x$?' Target prompt (creative): 'Write a short poem about the ocean' We measure KL divergence between the patched output distribution and the clean target output distribution. Lower KL = the patched output looks more like the source category's output, meaning that layer/feature is causally important.

8.2 Layer Sweep Results

- KL divergence decreases monotonically from layer 0 to layer 11.
- Layer 0: $KL \approx 3.5$ (patching early layers barely affects output)
- Layer 11: $KL \approx 0$ (patching the final layer almost completely overwrites the output)
- This monotonic decrease indicates that later layers are progressively more causally important for determining the model's output distribution.
- The most causally informative layer is layer 11 — the last layer before the unembedding.

8.2 Activation Patching — Layer Sweep

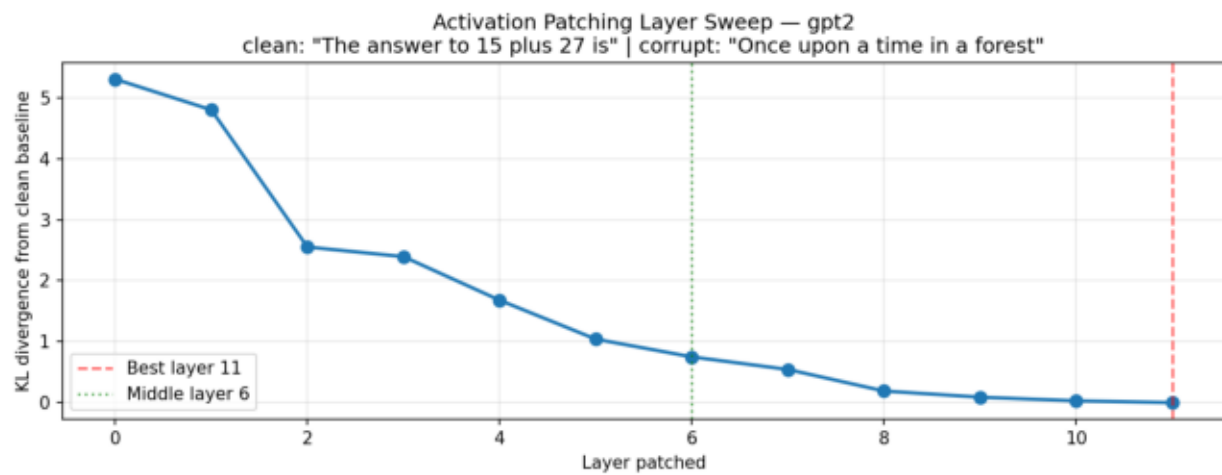


Figure 5: KL divergence from clean target output when patching each layer with source activations.

8.3 Surgical vs. Whole-Residual Patching

After identifying layer 11 as most causally important, we compared two patching strategies:

Whole-Residual Patch:

Replace the entire 768-dimensional residual stream at layer 11 with the source (math) activations. Result: $KL = 0.75$ (significant shift toward math-like output).

Surgical (Feature-Direction) Patch:

Replace only the top-20 discriminative math feature directions (identified in NB03) in the residual stream. Result: $KL = 3.38$ (barely moved from baseline $KL = 3.54$).

Interpretation:

- The surgical patch moved KL by only 0.16 (from 3.54 to 3.38), while the full patch moved it by 2.79 (from 3.54 to 0.75).
- This means the top-20 math features account for only ~6% of the total causal effect.
- Math-relevant information is DISTRIBUTED across many features, not concentrated in a few 'math neurons'.
- This is consistent with the superposition hypothesis: the model spreads each concept across many features in a high-dimensional space.
- To fully steer the model, you would need to patch hundreds of features, not just the top-20.

8.4 Feature Steering

We attempted to steer model output by amplifying the top-20 math feature directions (scaling by $5\times$) during generation from the creative prompt. The output was altered but not clearly math-like — further confirming that math behavior is distributed. More aggressive steering (higher multiplier, more features) or steering at multiple layers simultaneously might produce stronger effects.

9. Key Observations & Results Summary

► Prompt categories produce distinct feature signatures

Each category activates a measurably different set of SAE features, confirming that the model internally distinguishes between prompt types at the feature level.

► Early layers are most discriminative

Fisher's ratio peaks at layer 2 (1.33), suggesting the model quickly categorizes input type in early processing before blending representations for generation.

► Creative prompts are feature-rich, math is feature-sparse

Creative L0 ≈ 332 vs. math L0 ≈ 15 at layer 2. Creative writing demands broad conceptual activation; math is precise and narrow.

► Later layers are causally dominant

Activation patching shows monotonically increasing causal importance toward layer 11. The model makes its 'output decisions' in the final layers.

► Math information is distributed, not localized

Surgical patching of top-20 features captures only $\sim 6\%$ of the full patching effect. The model encodes math behavior across hundreds of features in superposition.

► Some features are shared across categories

Feature 22431 is top for both emotional and reasoning; feature 7467 for both math and code. This suggests higher-level abstract features (e.g., 'deliberation', 'formal language') span multiple categories.

► Factual category has the single strongest feature

Feature 6222 has a selectivity score of 10.25 for factual prompts — $2\times$ higher than any other category's top feature. Factual recall may be more localized than other capabilities.

10. Discriminative Feature Analysis

Top 5 most discriminative SAE feature indices per category (layer 6), ranked by selectivity score. Higher scores indicate stronger category specificity.

Category	Top-5 Feature Indices	Top Score
Math	12664, 10788, 7467, 8545, 6222	1.42
Code	15530, 17126, 7467, 8213, 9473	6.90
Factual	6222, 7484, 8545, 9525, 1753	10.25
Creative	24188, 21309, 8213, 17126, 10106	5.57
Emotional	22431, 8545, 7484, 6222, 19999	5.70
Reasoning	22431, 9525, 18217, 16247, 1753	5.22

Notable Patterns:

- Factual has the highest top-feature score (10.25) — factual recall is the most 'localized' capability.
- Code's top feature (15530, score 6.90) is highly exclusive — it barely fires for other categories.
- Math has the lowest top-feature score (1.42) — math is the most distributed capability.
- Shared features: 7467 (math+code), 8545 (math+factual+emotional), 1753 (code+factual+reasoning), 22431 (emotional+reasoning).
- Feature sharing patterns suggest a hierarchy: technical features (math/code), knowledge features (factual), and evaluative features (emotional/reasoning).

11. Reflection & Discussion

Q: Do different prompt types activate different features?

A: Yes, definitively. Each category has a distinct set of top-discriminative features with minimal overlap. The cosine similarity analysis shows clear separation at early layers, and the feature heatmap reveals highly category-specific activation patterns. This confirms that LLMs don't process all text identically — they develop specialized internal circuits for different types of reasoning.

Q: At which layer do features best separate prompt types?

A: Layer 2 (early) provides the best separability with a Fisher ratio of 1.33, dropping to ~0.6 at layers 6 and 10. This is a counterintuitive finding — one might expect deeper layers to be more semantically rich. However, it suggests that the model performs rapid input categorization in early layers, then progressively transforms representations toward a universal generation format.

Q: What does sparsity tell us about how the model processes different inputs?

A: Sparsity (L0) varies dramatically by category. Creative prompts activate 20× more features than math prompts at layer 2. This reflects the inherent complexity difference: creative writing draws on diverse concepts (metaphor, imagery, emotion, structure), while math requires precise activation of specific computational features. The model mirrors human intuition — creative thinking is 'broader', analytical thinking is 'narrower'.

Q: Can we causally verify that features drive behavior?

A: Partially. Whole-residual patching at layer 11 strongly shifts output (KL drops from 3.54 to 0.75), confirming that layer carries causal information. However, surgical patching of the top-20 features only accounts for ~6% of this effect. This means we've identified the right features (they are correlated with categories), but the causal mechanism is distributed. The model doesn't have 'math neurons' — it has a distributed math representation spread across hundreds of features.

Q: What are the limitations of this study?

A: (1) We used GPT-2 Small due to hardware constraints — results may differ for larger models. (2) Last-token activation captures only the final representation, missing how features evolve across token positions. (3) SAE features are an approximation of the model's true features — SAE training introduces its own biases. (4) Our prompt dataset (120 prompts) is relatively small — more prompts would improve statistical power. (5) Feature interpretation relies on Neuronpedia labels, which may be incomplete or inaccurate.

12. Conclusions & Future Work

Conclusions:

- LLMs develop category-specific internal representations that are detectable through SAE feature analysis. Different prompt types genuinely activate different features.
- Feature separability peaks at early layers (layer 2), suggesting rapid input categorization followed by progressive representation blending toward generation.
- Creative writing is the most feature-rich task (highest L0), while math is the most sparse — reflecting the breadth vs. precision tradeoff.
- Causal verification through activation patching confirms that layer 11 is most causally important, but reveals that math-relevant information is distributed across hundreds of features rather than localized in a few.
- This distribution pattern supports the superposition hypothesis: the model encodes many more concepts than it has dimensions by spreading each concept across multiple features.

Future Work:

- Scale to larger models (Gemma 2 2B, 9B) to see if feature localization improves with model capacity.
- Analyze feature evolution across token positions, not just last-token activations.
- Use more aggressive feature steering (100+ features, multi-layer intervention, higher multipliers).
- Expand the prompt dataset to 500+ prompts per category for stronger statistical conclusions.
- Cross-reference with Neuronpedia annotations to build a semantic map of discovered features.
- Investigate whether category-specific features correspond to attention head specialization.
- Apply similar analysis to instruction-tuned models to see how fine-tuning reorganizes feature representations.

End of Report

Model: GPT2 • SAE: gpt2-small-res-jb • 24,576 features • Layers [2, 6, 10]

CS 463 — Machine Learning — Spring 2025